

Question	Marking details	Marks Available																	
		AO1	AO2	AO3	Total	Maths	Prac												
6	Indicative content <u>Structure of water:</u> Water has the chemical formula of H ₂ O. Water consists of one oxygen atom joined to two hydrogen atoms by covalent bonds. The molecule is a dipole. The oxygen atom has a slight negative charge / δ-, the hydrogen atoms have a slightly positive charge δ+.																		
	<table><tr><th><u>Properties of water</u></th><th><u>Essential to supporting life</u></th></tr><tr><td>Ice is less dense than water</td><td>Ice floats insulating the water below allowing organisms to survive.</td></tr><tr><td>Polar solvent</td><td>All chemical reactions occur in aqueous solutions. Polar molecules and ions can dissolve and therefore be transported</td></tr><tr><td>Metabolite</td><td>Involved in chemical reactions</td></tr><tr><td>High specific heat capacity</td><td>Requires a lot of energy to cause water to heat up / cool down therefore creating thermostable environments</td></tr><tr><td>Cohesive / forms hydrogen bonds</td><td>Water molecules form hydrogen bonds allowing the movement of water</td></tr></table>	<u>Properties of water</u>	<u>Essential to supporting life</u>	Ice is less dense than water	Ice floats insulating the water below allowing organisms to survive.	Polar solvent	All chemical reactions occur in aqueous solutions. Polar molecules and ions can dissolve and therefore be transported	Metabolite	Involved in chemical reactions	High specific heat capacity	Requires a lot of energy to cause water to heat up / cool down therefore creating thermostable environments	Cohesive / forms hydrogen bonds	Water molecules form hydrogen bonds allowing the movement of water						
	<u>Properties of water</u>	<u>Essential to supporting life</u>																	
	Ice is less dense than water	Ice floats insulating the water below allowing organisms to survive.																	
	Polar solvent	All chemical reactions occur in aqueous solutions. Polar molecules and ions can dissolve and therefore be transported																	
	Metabolite	Involved in chemical reactions																	
	High specific heat capacity	Requires a lot of energy to cause water to heat up / cool down therefore creating thermostable environments																	
	Cohesive / forms hydrogen bonds	Water molecules form hydrogen bonds allowing the movement of water																	
	<u>Chemical elements:</u> Phosphorous is required to synthesise phospholipids / nucleotides / ATP / DNA / RNA Sulfur is required to synthesise some amino acids / methionine / cysteine / proteins.																		